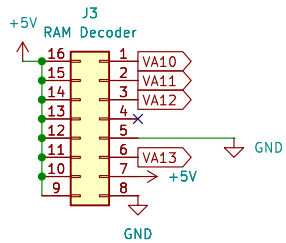
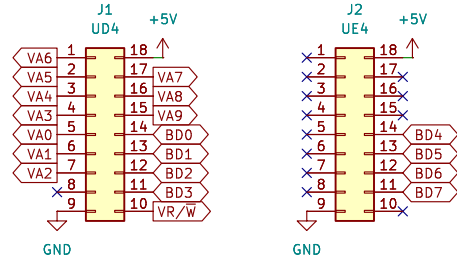


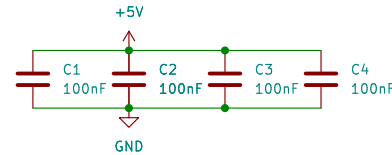
Signals to/from RAM decoder (74LS138)
Original decoder activates internal RAM only for RAM0 and RAM4 to RAM5
Now greatly simplified – we essentially don't need to care about the RAMx signals at all, just pull all the original 74LS138 outputs high.
But we do need to use the incoming signals.



Original 2114 RAM chips are removed from the VIC.
Two are replaced with IC sockets so daughterboard can be installed.

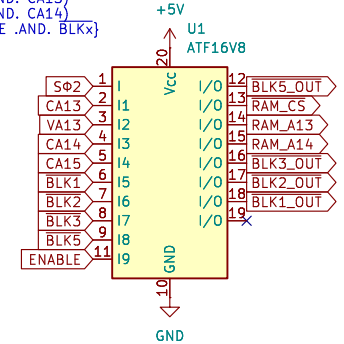


Power & Decoupling



Remove lots of different logic gates with a single PLD
We need to handle whether CPU or VIC needs to access the RAM
When SΦ2 is low, CPU is accessing RAM and CA13/CA14 are valid
When SΦ2 is high, VIC is accessing RAM and CA13/CA14 must be low as VIC only reads BLK0
If expansion is disabled then route BLK1/BLK2/BLK3/BLK5 back to motherboard

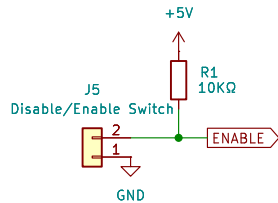
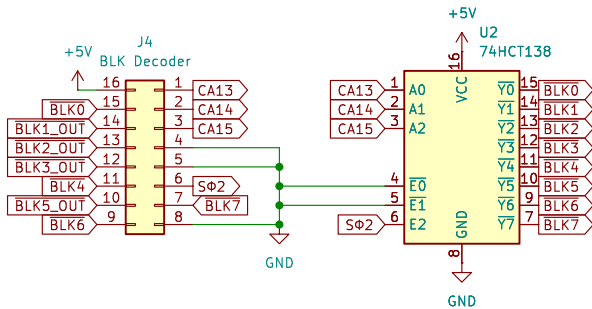
RAM_CS = (SΦ2 .AND. CA15) .OR. (SΦ2 .AND. VA13)
RAM_A13 = (SΦ2 .AND. CA13)
RAM_A14 = (SΦ2 .AND. CA14)
BLKx_OUT = (ENABLE .AND. BLKx)



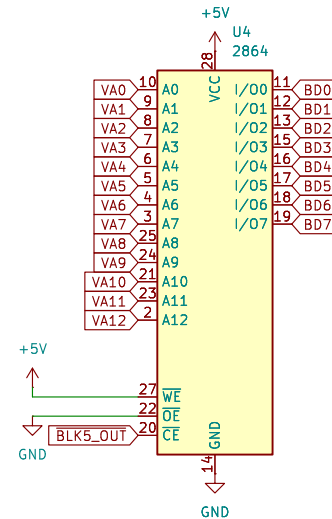
Signals to/from BLK decoder (74LS138)

When our expansion is active we want to prevent BLK1/BLK2/BLK3/BLK5 selects returning to the VIC
But we also want to be able to disable the expansion for software compatibility
This can be done via a switch:

* When the expansion is disabled the BLK1/BLK2/BLK3/BLK5 are passed back to the VIC, disconnecting our RAM expansion from the data bus and passing through to the cartridge port
* When the expansion is enabled, these are blocked which prevents the VIC from disconnecting our RAM from the data bus

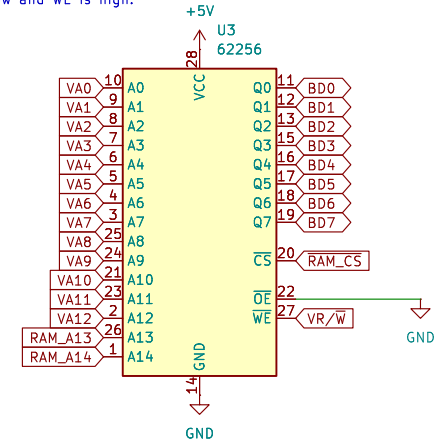


8KB ROM Expansion
Sneak in a ROM, like SUPER EXPANDER, in BLK5 (0xA000)



32KB RAM Expansion

Read: CS low & VR/W high → OE low
Write: CS low & VR/W low → OE high
Dave's comment: The internal logic will take over if OE is tied low. It will only output if CS is low and WE is high.



With lots of help from Dave Curran @ Tynemouth Software

Refer to the "DIP-16 Breakout Board" for the RAM/BLK Decoder connection

Brett Hallen

Sheet: /

File: Internal_32KB_RAM_8KB_ROM.kicad_sch

Title: Commodore VIC-20 32KB RAM+ROM Expansion

Size: A4 Date: 12/JUN/2026

KiCad E.D.A. 9.0.9

Rev: A

Id: 1/1